PRODUCT USER MANUAL

For Regional High Resolution Sea Ice Charts Svalbard and Greenland Region

SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_002

Issue: 3.0

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2.8	2019.12.06	All	Greenland datasets (former _011_003) product) information included.	Matilde B. Kreiner	C. Wettre
2.9	2020.09.09	All	Updated with information of new Greenland datasets.	Matilde B. Kreiner	C. Wettre
3.0	2024.09.01	I, VI	Updates of the Svalbard dataset	Frode Dinessen	C. Wettre

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GLOSSARY AND ABBREVIATIONS

TAC	Thematic Assembly Centre
SI	Sea Ice
MET Norway	Norwegian Meteorological Institute
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
DMI	Danish Meteorological Institute
RMS	Root mean square
SDN	SeaDataNet (climatology)
PC	Production Center
PU	Production Unit
Sigrid-3	Sea Ice GeoReferenced Information and Data, version 3
Directgetfile	CMEMS service tool (FTP like) to download a NetCDF file
WMO	World Meteorological Organization

DATA ACCES

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1 Summary

This guide describes the regional sea ice product SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS _011_002 from the CMEMS Sea Ice Thematic Ensemble Centre (SI TAC), what data services are available, how to access them and how to use the files and services.

This products datasets are produced at MET Norway and DMI and disseminated by MET Norway. They are high resolution sea ice chart products covering the regions around Greenland, around Svalbard and from the Barents Sea region to Novaya Zemlya.

The data are freely available under the CMEMS data license.

I.2 History of changes

- **Since October 2014**, Sentinel-1(A) data has gradually been incorporated into the service and this satellite has now replaced most of the RADARSAT data.
- **Since June 2016**, Sentinel-1B data have been incorporated into the service. Both Sentinel-1A and -1B are now fully integrated into the product production.
- **Since November 2024**, Due to a new and updated production chain, the ice chart dataset from MET Norway, METNO-ARC-SEAICE_CONC-L4-NRT-OBS, has been replaced by a new dataset, cmems_obs_si_arc_phy_nrt_1km_svb_P1D-irr. The main changes are:

1. The projection parameter of the netCDF file has been updated with well-known text representation of coordinate reference systems. This also includes adding "+ellps=sphere" to the proj4 string. Files are updated to netCDF convention CF-1.10
2. A spatial offset of the sea ice concentration has been fixed.
3. The algorithm used to convert ice polygons from vector to grid has changed to more closely follow the polygon boundaries.
4. A data quality variable has been added which is a sum of available satellite images on the day the chart was produced where Sentinel-1 = 1, Radarsat-2 = 2, Cosmo SkyMed = 4.
5. The dataset concentration_range has been removed.
6. New and updated metadata

   All products delivered to CMEMS have been reprocessed and updated to follow the new format netCDF format.

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Description

The operational sea ice services at MET Norway and at DMI produces sea ice concentration charts based on a manual interpretation of satellite data.

The satellite data used are Synthetic Aperture Radar (SAR) data from Envisat (until its failure in April 2012) and Radarsat. Since October 2014, Sentinel-1 data has gradually been incorporated into the service and is now the dominant input data source. However, SAR data from Copernicus contributing missions are also extensible used. In addition to the SAR data, visual and infrared data from Sentinel-2, Sentinel-3, VIIRS NOAA-AVHRR and passive microwave data from AMSR2, are used.

The Sentinel-1 data are in Extended Wide (EW) mode dual polarization. Sentinel-1 data in Interferometric Wide swath mode (IW) is also used when available. Each EW scene covers an area of 400 x 400 km and is resampled to a spatial horizontal resolution of 80x80 meter. An IW scene covers an area of 250 x 250 km. The Sentinel-1 data covers most of the MET Norway and DMI ice charting areas. The Radarsat data are in Scansar Wide mode dual polarization. Each scene covers an area of 500 x 500 km and is resampled to a spatial horizontal resolution of 50x50 meter. The VIIRS and NOAA data cover the charting area several times each day and are resampled to a horizontal resolution of 1000 x 1000 meter.

The operator uses the latest available satellite data and draws the ice chart in a QGIS production system.

The Svalbard and Greenland datasets have different temporal frequency: The Svalbard dataset is distributed on working days before 15.00 UTC. The Greenland overview ice chart dataset is produced twice weekly and summarizes ice conditions based on satellite data acquired within the last 1-3 days. The Greenland regional ice chart dataset coverage and frequency depends on season, ship routes and data availability. Normally one regional ice chart is produced on the basis of one Sentinel-1 image and the individual scenes/ice charts cover a subset of the DMI ice charting area of Greenland waters. The ice charts are thus snapshot interpretations of the ice conditions at the time of the Sentinel-1 image acquisition.

All the gridded sea ice chart products are available at a spatial resolution of 1000 meter.

II.2 General Information

Product Lines	SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_002
Geographical coverage	Svalbard area 80°W-80°E ; 60°N-85°N, Greenland area 80°W-10°E, 55°N-85°N

Variables	Svalbard dataset: Total ice concentration, quality variable Greenland datasets: total and partial ice concentrations, stage of development, floe size.
Analysis	Yes
Forecast	No
Available time series	Svalbard ice chart: From 4 January 2010 – on going Greenland regional ice charts: From 5 January 2009 – ongoing Greenland overview ice charts: From 2 March 2011 – ongoing
Temporal resolution	Svalbard ice charts: daily on weekdays. Greenland regional ice charts: When ice charts are produced (0 - more times per day). Greenland overview ice charts: produced on Sundays and Wednesdays.
Target delivery time	Svalbard ice charts: Daily on working days before 15 UTC Greenland regional ice charts: Within 18 hours after satellite data acquisition. Greenland overview ice charts: 14:00 UTC next day (Mondays and Thursdays).
Horizontal resolution	Svalbard ice chart: 1km Greenland ice charts: ~1-5km, gridded at 1 km grid
Number of vertical levels	Surface parameter
Format	NetCDF CF1.4

Table 1 - SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_002 Product Specification

Detailed information on the systems and products are on CMEMS web site: http://marine.copernicus.eu/

II.3 Details of datasets

The MET Norway and DMI ice chart datasets follow the World Meteorological Organization (WMO) code for sea ice. The Svalbard and Greenland datasets are listed in the table below.

DATASETS	VARIABLES AND UNIT	NAME OF VARIABLES IN THE NETCDF FILE
cmems_obs_si_arc_phy_nrt_1km_svb_P1D-irr	Ice concentration [%]	"sea_ice_area_fraction"
	Data quality given as sum of available satellite data used in production.	“quality flag”
cmems_obs-si_arc_physic_nrt_1km-grl_P1WT3D-m	Ice concentration [%]	"sea_ice_area_fraction"
	Concentration range [%]	“concentration_range”
	Reference ids for rasterized sea-ice-polygons	“ice_poly_id_grid”
	IDs corresponding to sea-ice-polygon reference IDs	“polygon_id”
	Total concentration*	“CT”
	Partial concentration of thickest ice*	“CA”
	Partial concentration of second thickest ice*	“CB”
	Partial concentration of the third thickest ice*	“CC”
	Stage of development of thickest ice*	“SA”
	Stage of development of second thickest Ice*	“SB”
	Stage of development of third thickest ice*	“SC”
	Form of thickest ice*	“FA”
	Form of second thickest ice*	“FB”
	Form of third thickest ice*	“FC”
	Stage of development of ice thicker than SA but with concentration less than 1/10*	“CN”
	Stage of development of any remaining class of ice*	“CD”
	Predominant and secondary forms of ice*	“CF“
	General polygon surface type [I=Ice, W=Water]	“poly_type”
	* Coded in Sigrid-3. See Table 3 - Table 6.	

cmems_obs-si_arc_physic_nrt_1km-grl_P1D-irr	Ice concentration [%]	"sea_ice_area_fraction"
	Concentration range [%]	"concentration_range"
	Reference ids for rasterized sea-ice-polygons	"ice_poly_id_grid"
	IDs corresponding to sea-ice-polygon reference IDs	"polygon_id"
	Total concentration*	"CT"
	Partial concentration of thickest ice*	"CA"
	Partial concentration of second thickest ice*	"CB"
	Partial concentration of the third thickest ice*	"CC"
	Stage of development of thickest ice*	"SA"
	Stage of development of second thickest Ice*	"SB"
	Stage of development of third thickest ice*	"SC"
	Form of thickest ice*	"FA"
	Form of second thickest ice*	"FB"
	Form of third thickest ice*	"FC"
	Stage of development of ice thicker than SA but with concentration less than 1/10*	"CN"
	Stage of development of any remaining class of ice*	"CD"
	Predominant and secondary forms of ice*	"CF"
	General polygon surface type [I=Ice, W=Water]	"poly_type"
	* Coded in Sigrid-3. See Table 3 - Table 6.	

Table 2 - List of the datasets

Svalbard dataset:

For the Svalbard dataset, cmems_obs_si_arc_phy_nrt_1km_svb_P1D-irr, the concentration classes are defined by concentration intervals:

- *Fast ice: 10/10*
- *Very Close Drift Ice: 9-10/10*
- *Close Drift Ice: 7-8/10*
- *Open Drift Ice: 4-6/10*
- *Very Open Drift Ice: 1-3/10*
- *Open Water: < 1/10*

These concentration intervals are converted to fixed concentration values in the gridded product described by the "sea_ice_area_fraction" variable:

- *Fast ice (10/10th)s: 100*
- *Very Close Drift Ice (9-10/10ths): 95*
- *Close Drift Ice (7-9/10ths): 80*
- *Open Drift Ice (4-7/10ths): 55*
- *Very Open Drift Ice (1-4/10ths): 25*
- *Open Water (0-1/10ths): 5*
- *Ice Free: 0*

Greenland datasets:

The DMI datasets follow the World Meteorological Organization (WMO) code for sea ice with concentration classes defined by concentrations in 10ths, and the total ice concentration variable *sea_ice_are_fraction* gives concentration in 10ths, from 0 (open water) to 100% ice (landfast ice). To reflect the concentration intervals the *concentration_range* variable is given in the product defined by a +- concentration value for each grid point. Concentration intervals are now rarely used in the DMI ice charts.

Note: The ice concentrations are seldom homogeneously distributed inside the drawn polygons down to the scale of the gridding, so whereas the average concentrations of the polygons are considered correct for the polygon, each individual 1 x 1 km grid cell may deviate substantially from this average value.

In addition to the total ice concentration variable *sea_ice_are_fraction* and the concentration interval variable *concentration_range* the DMI datasets contain all additional sea ice variables from the WMO egg code, see Figure 1, in SIGRID-3 code from the DMI overview and regional ice charts. The SIGRID-3 ice codes consists of a total ice concentration for the polygon (CT), concentrations of up to 3 different ice types/forms specified by their partial concentration (CA, CB and CC), their stage of development (SA, SB and SC) and their form (FA, FB and FC). These are explained in the following Table 3 - Table 6 (extracted from JCOMM Expert Team on Sea Ice, 2014).

When converting the original ice chart format into the dataset netCDF format, each polygon is given an ID number, see variable *ice_poly_id_grid* and *polygon_id*, and all ice chart variables (*CT, CA, CB, CC, SA,* …) for that polygon is described in the netCDF file organized by this ID number.

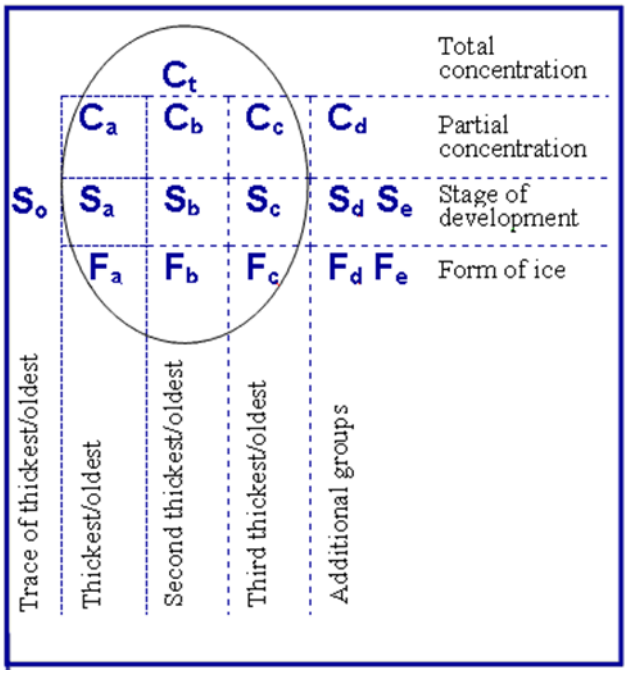

Figure 1 - Overview of ice egg code variables: The World Meteorology Organization (WMO) system for sea ice symbology is referred to as the "Egg Code" due to the oval shape of the symbol. Overview of ice egg code variables - See also code tables later in this section. (Source: nsidc.org).

Definition, concentration	Sigrid3 code (CT, CA, CB and CC)
Ice Free	00
Less than 1/10 (open water)	01
Bergy water	02
1/10	10
2/10	20
3/10	30
4/10	40
5/10	50
6/10	60
7/10	70
8/10	80
9/10	90
10/10	92

Table 3 - Concentration codes for variable identifiers CT, CA, CB, and CC (Specific concentrations).

Stage of Development	Thickness	Sigrid3 code for variables SA, SB, SC, CN and CD
Ice Free		00
No stage of development		80
New Ice	< 10 cm	81
Nilas, Ice Rind	< 10 cm	82
Young Ice	10 - 30 cm	83
Grey Ice	10 - 15 cm	84
Grey – White Ice	15 - 30 cm	85
First Year Ice	30 - 200 cm	86
Thin First Year Ice	30 - 70 cm	87
Thin First Year Ice Stage 1	30 - 50 cm	88
Thin First Year Ice Stage 2	50 - 70 cm	89
Medium First Year Ice	70 - 120 cm	91
Thick First Year Ice	> 120 cm	93
Old Ice		95
Second Year Ice		96
Multi-Year Ice		97
Glacier Ice		98

Table 4 - Stage of development codes and thickness of ice for variable identifiers SA, SB, SC, CN, and CD.

Form	Size/ concentration	Sigrid3 code for variables FA, FB, FC and CF
Pancake Ice	30 cm – 3 m	00
Shuga / Small Ice Cake, Brash Ice	< 2 m across	01
Ice Cake	< 20 m across	02
Small Floe	20 – 100 m across	03
Medium Floe	100 – 500 m across	04
Big Floe	500 m – 2 km across	05
Vast Floe	2 – 10 km across	06

Giant Floe	> 10 km across	07
Fast Ice		08
Growlers, Floebergs or Floebits		09
Icebergs		10
Strips and Patches	Concentrations 1/10	11
Strips and Patches	Concentrations 2/10	12
Strips and Patches	Concentrations 3/10	13
Strips and Patches	Concentrations 4/10	14
Strips and Patches	Concentrations 5/10	15
Strips and Patches	Concentrations 6/10	16
Strips and Patches	Concentrations 7/10	17
Strips and Patches	Concentrations 8/10	18
Strips and Patches	Concentrations 9/10	19
Strips and Patches	Concentrations 10/10	20
Level Ice		21
Unknown		99

Table 5 - Form of ice codes for variable identifiers FA, FB, FC, and CF.

Description	poly_type
Water – sea ice free	L
Ice – of any concentration	W
No Data	N

Table 6 - List of the variable Poly_type character variables.

II.4 Additional information about products

The ice charts are originally produced for navigational purposes and consists of manually drawn polygons with fairly homogenous ice conditions. The ice charts contains a number of ice parameters, of which the total ice concentration is dominant. The ice concentrations are not always homogenously distributed inside the polygons down to the 1 kilometer scale of the gridding, so whereas the average concentrations of the polygons are considered correct for the polygon, each individual 1km grid cell may deviate substantially from this average value.

III FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:

What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV NOMENCLATURE OF FILES

The nomenclature of the downloaded files differs on the basis of the chosen download mechanism **Subset**

File name convention:

ice_conc_svalbard_<yyyymmddhhmm>.nc
ice_conc_greenland_<yyyymmddhhmm>.nc
ice_conc_overview_greenland _<yyyymmddhhmm>.nc

The file naming convention is relevant when downloading by **Marine Data Store**.

IV.1 Nomenclature of files when downloaded through the CMEMS Web Portal Subset Service

Files nomenclature when downloaded through the CMEMS Web Portal Subset is based on product dataset name and a numerical reference related to the request date on the MOTU.

The scheme is: **datasetname-nnnnnn.nc**

where:

.datasetname is a character string within one of the following:

· ice_conc_svalbard_aggregated

· cmems_obs-si_arc_physic_nrt_1km-grl_P1WT3D-m

· cmems_obs-si_arc_physic_nrt_1km-grl_P1D-irr

. nnnnnn: 6 digit transaction identifier corresponding to the unique request number in MOTU.

.nc: standard NetCDF filename extension.

Example:

`ice_conc_svalbard_aggregated_130346.nc`

IV.2 Nomenclature of original files

When downloading through Marine Data Store you will get a file named **datasetname_nnnnnn.zip** where the datasetname and nnnnnn are as explained for Subset.

The files in the Directgetfile zip package follow the file naming convention shown at the beginning of Ch. IV.

IV.3 Grid type and Geographical Projection

The product datasets are gridded in a regular grid of 1000x1000 meters covering the areas shown in the figures below. The projection used is polar-stereographic with the following projection parameters:

Polarstereographic (+proj=stere +lat_0=90n +lon_0=0e +lat_ts=90n +ellps=sphere +A=6371000.0)

IV.4 Domain coverage (includes image of domain + projection used if any)

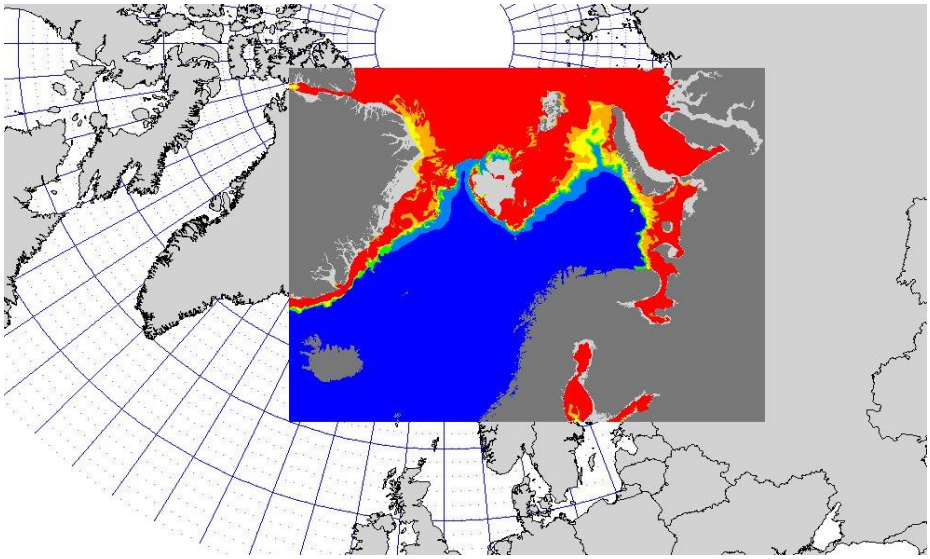

Figure 2 - Example of full Svalbard dataset spatial coverage (produced on every weekday).

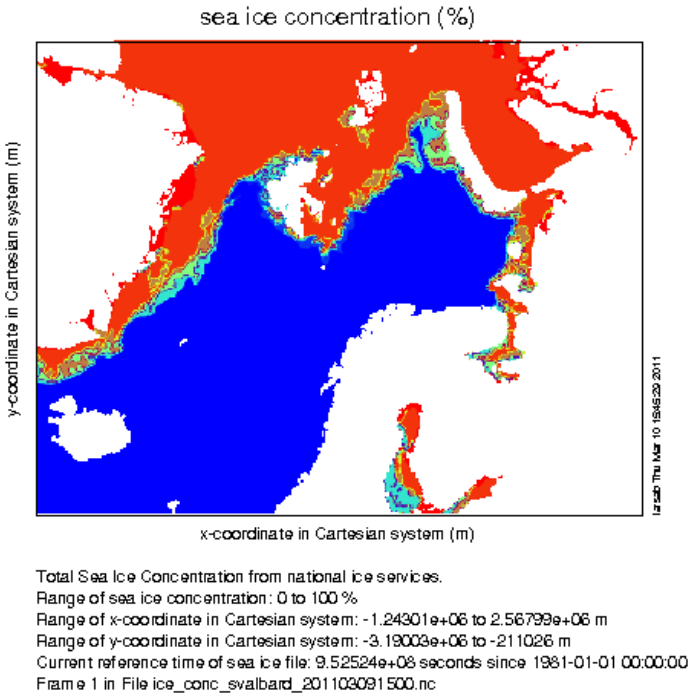

Figure 3 - Spatial coverage of the cmems_obs_si_arc_phy_nrt_1km_svb_P1D-irr product

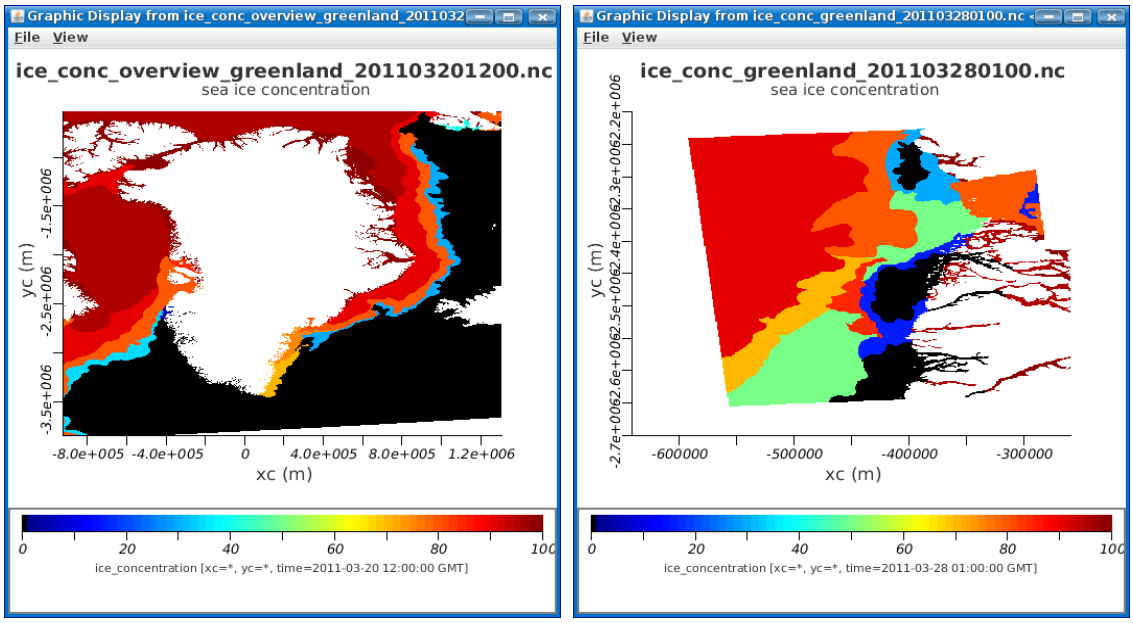

Figure 4 - Spatial coverage of the Greenland datasets. Right: Overview datasets fixed coverage (produced twice a week). Left: Example of regional 'NRT' dataset coverage

(produced at varying frequencies for different areas).

IV.5 Update Time

Svalbard dataset:

- updated every weekday at 4 pm.

Greenland datasets:

- *Regional* ice chart dataset: updated up to several times every weekday within 18 hrs of satellite data acquisition.
- *Overview* ice chart dataset: updated every Monday and Thursday by 14:00 UTC.

The Greenland datasets may not be updated on public Danish holidays, but updated the day after.

IV.6 Temporal extent of analysis and forecast stored on delivery mechanism

Svalbard dataset:

stored since 1. January 2010.

Greenland datasets:

- Regional ice chart dataset, cmems_obs-si_arc_physic_nrt_1km-grl_P1D-irr: available from 15 December 2020 -.
- Overview ice chart dataset, cmems_obs-si_arc_physic_nrt_1km-grl_P1WT3D-m: available from 15 December 2020 -.
- *Regional ice chart dataset, DMI-ARC-SEAICE_CONC-L4-NRT-OBS: available from 5 January 2009 – 15 February 2021.*
- *Overview ice chart dataset, DMI-ARC-SEAICE_CONC_OVERVIEW-L4-NRT-OBS: available from 3 March 2011 – 15 February 2021.*

IV.7 Other information: mean centre of Products, land mask value, missing value

Land mask values are defined as value -99

IV.8 Structure of files

Examples of the header of output NetCDF files are inserted in annex, for each dataset.

V ANNEX

Svalbard dataset, cmems_obs_si_arc_phy_nrt_1km_svb_P1D-irr:

Svalbard dataset, cmems_obs_si_arc_phy_nrt_1km_svb_P1D-irr.:

dimensions:

time = 1 ;

xc = 3812 ;

yc = 2980 ;

variables:

int time(time) ;

time:units = "seconds since 1970-01-01 00:00:00 +00:00" ;

time:standard_name = "time" ;

time:long_name = "time" ;

time:calendar = "proleptic_gregorian" ;

time:axis = "T" ;

double xc(xc) ;

xc:units = "meters" ;

xc:standard_name = "projection_x_coordinate" ;

xc:axis = "X" ;

double yc(yc) ;

yc:units = "meters" ;

yc:standard_name = "projection_y_coordinate" ;

yc:axis = "Y" ;

float ice_concentration(time, yc, xc) ;

ice_concentration:_FillValue = -99.f ;

ice_concentration:long_name = "Sea ice concentration. 0: Ice Free (CT98), 5: Open Water (CT01), 25: Very Open Drift Ice (CT14), 55: Open Drift Ice (CT47), 80: Close Drift Ice (CT79), 95: Very Close Drift Ice (CT91), 100: Fast Ice (CT92, FP08)" ;

ice_concentration:standard_name = "sea_ice_area_fraction" ;

ice_concentration:units = "%" ;

ice_concentration:date = "2022-12-12T15:00:00" ;

ice_concentration:coordinates = "lat lon" ;

ice_concentration:grid_mapping = "crs" ;

ice_concentration:source = "MET Norway" ;

ice_concentration:ancillary_variables = "satellite_availability" ;

double crs ;

crs:_FillValue = NaN ;

crs:crs_wkt = "PROJCRS[\"unknown\",BASEGEOGCRS[\"unknown\",DATUM[\"Unknown based on Normal Sphere (r=6370997) ellipsoid\",ELLIPSOID[\"Normal Sphere (r=6370997)\",6370997,0,LENGTHUNIT[\"metre\",1,ID[\"EPSG\",9001]]]],PRIMEM[\"Greenwich\",0,ANGLEUNIT[\"degree\",0.0174532925199433],ID[\"EPSG\",8901]]],CONVERSION[\"unknown\",METHOD[\"Polar Stereographic (variant B)\",ID[\"EPSG\",9829]],PARAMETER[\"Latitude of standard parallel\",90,ANGLEUNIT[\"degree\",0.0174532925199433],ID[\"EPSG\",8832]],PARAMETER[\"Longitude of origin\",0,ANGLEUNIT[\"degree\",0.0174532925199433],ID[\"EPSG\",8833]],PARAMETER[\"False easting\",0,LENGTHUNIT[\"metre\",1],ID[\"EPSG\",8806]],PARAMETER[\"False northing\",0,LENGTHUNIT[\"metre\",1],ID[\"EPSG\",8807]]],CS[Cartesian,2],AXIS[\"(E)\",south,MERIDIAN[90,ANGLEUNIT[\"degree\",0.0174532925199433,ID[\"EPSG\",9122]]],ORDER[1],LENGTHUNIT[\"metre\",1,ID[\"EPSG\",9001]]],AXIS[\"(N)\",south,MERIDIAN[180,ANGLEUNIT[\"degree\",0.0174532925199433,ID[\"EPSG\",9122]]],ORDER[2],LENGTHUNIT[\"metre\",1,ID[\"EPSG\",9001]]]]" ;

crs:proj4_string = "+proj=stere +lat_0=90 +lat_ts=90 +lon_0=0 +x_0=0 +y_0=0 +ellps=sphere +units=m +no_defs +type=crs" ;

crs:semi_major_axis = 6370997. ;

crs:semi_minor_axis = 6370997. ;

crs:inverse_flattening = 0. ;

crs:reference_ellipsoid_name = "Normal Sphere (r=6370997)" ;

crs:longitude_of_prime_meridian = 0. ;

crs:prime_meridian_name = "Greenwich" ;

crs:geographic_crs_name = "unknown" ;

crs:horizontal_datum_name = "Unknown based on Normal Sphere (r=6370997) ellipsoid" ;

crs:projected_crs_name = "unknown" ;

crs:grid_mapping_name = "polar_stereographic" ;

crs:standard_parallel = 90. ;

crs:straight_vertical_longitude_from_pole = 0. ;

crs:false_easting = 0. ;

crs:false_northing = 0. ;

float lat(yc, xc) ;

```
            lat:grid_mapping = "crs" ;
            lat:units = "degrees_north" ;
            lat:standard_name = "latitude" ;
            lat:long_name = "latitude" ;
        float lon(yc, xc) ;
            lon:grid_mapping = "crs" ;
            lon:units = "degrees_east" ;
            lon:standard_name = "longitude" ;
            lon:long_name = "longitude" ;
        int satellite_availability(time, yc, xc) ;
            satellite_availability:_FillValue = -99 ;
            satellite_availability:long_name = "Sum of available satellite data. Sentinel-1: 1, Radarsat-2: 2, Cosmo SkyMed: 4" ;
            satellite_availability:standard_name = "quality_flag" ;
            satellite_availability:date = "2022-12-12T15:00:00" ;
            satellite_availability:grid_mapping = "crs" ;
            satellite_availability:source = "MET Norway" ;
            satellite_availability:flag_values = 1, 2, 3, 4, 5, 6, 7 ;
            satellite_availability:flag_meanings = "Sentinel1 Radarsat2 Sentinel1_Radarsat2 CosmoSkyMed CosmoSkyMed_Sentinel1 CosmoSkyMed_Radarsat2 CosmoSkyMed_Radarsat2_Sentinel1 " ;

// global attributes:
            :id = "3bf9e215-4c41-462c-990f-e3b82492f4c6" ;
            :naming_authority = "no.met" ;
            :date_created = "2024-05-14T11:18:33+00:00" ;
            :creator_institution = "Norwegian Meteorological Institute" ;
            :institution = "Norwegian Meteorological Institute" ;
            :institution_short_name = "MET Norway" ;
            :title = "Arctic Sea Ice Concentration From Norwegian Meteorological Institute" ;
            :title_no = "Arktisk sjøiskonsentrasjon Fra Meteorologisk Institutt" ;
            :summary = "The ice concentration is based on a manual interpretation of satellite data. The landmask data are from the Norwegian Polar Institute (Svalbard) and OpenStreetMap (all other land areas)." ;
```

:summary_no = "Iskonsentrasjonen er basert på en manuell tolkning av satellittdata. Landmaskedataene er fra Norsk Polarinstitutt (Svalbard) og OpenStreetMap (alle andre landområder)." ;

:keywords = "GCMDSK:EARTH SCIENCE > CRYOSPHERE > SEA ICE > SEA ICE CONCENTRATION > ICE FRACTION, GEMET:Sea Regions, GEMET:Oceanographic geographical features, NORTHEMES:Weather and climate" ;

:keywords_vocabulary = "GCMDSK:GCMD Science Keywords:https://gcmd.earthdata.nasa.gov/kms/concepts/concept_scheme/sciencekeywords, GEMET:INSPIRE themes:https://inspire.ec.europa.eu/theme, NORTHEMES:GeoNorge Themes:https://register.geonorge.no/subregister/metadata-kodelister/kartverket/nasjonal-temainndeling" ;

:iso_topic_category = "climatologyMeteorologyAtmosphere" ;

:license = "http://spdx.org/licenses/CC-BY-4.0(CC-BY-4.0)" ;

:publisher_name = "Norwegian Meteorological Institute" ;

:publisher_type = "institution" ;

:publisher_email = "istjenesten@met.no" ;

:publisher_url = "https://www.met.no/" ;

:Conventions = "CF-1.10, ACDD-1.3" ;

:history = "Python Xarray" ;

:time_coverage_start = "2022-12-12T15:00:00+00:00" ;

:time_coverage_end = "2022-12-12T15:00:00+00:00" ;

:spatial_representation = "grid" ;

:geospatial_lat_max = 90LL ;

:geospatial_lat_min = 60LL ;

:geospatial_lon_max = 85LL ;

:geospatial_lon_min = -80LL ;

:processing_level = "Experimental" ;

}

Greenland dataset: cmems_obs-si_arc_physic_nrt_1km-grl_P1WT3D-m, example:

```
netcdf file:ice_conc_overview_greenland_202009061200.nc {
 dimensions:
   time = 1;
   yc = 3318;
   xc = 2232;
   polygon_reference = 1000;
```

```
variables:
  int time(time=1);
    :long_name = "reference time of sea ice file";
    :units = "seconds since 1981-01-01 00:00:00";
    :calendar = "standard";

  float yc(yc=3318);
    :axis = "Y";
    :long_name = "y-coordinate in Cartesian system";
    :units = "m";

  float xc(xc=2232);
    :axis = "X";
    :long_name = "x-coordinate in Cartesian system";
    :units = "m";

  float lat(yc=3318, xc=2232);
    :long_name = "latitude";
    :units = "degrees_north";
    :_ChunkSizes = 1106U, 744U; // uint

  float lon(yc=3318, xc=2232);
    :long_name = "longitude";
    :units = "degrees_east";
    :_ChunkSizes = 1106U, 744U; // uint

  char crs;
    :straight_vertical_longitude_from_pole = -45L; // long
    :latitude_of_projection_origin = 90L; // long
    :standard_parallel = 90L; // long
    :grid_mapping_name = "polar_stereographic";
    :false_easting = 0.0f; // float
    :false_northing = 0.0f; // float
    :proj4_string = " +proj=stere +lon_0=-45 +lat_ts=90 +lat_0=90 +a=6371000 +b=6371000";

  short ice_concentration(time=1, yc=3318, xc=2232);
    :long_name = "sea ice concentration";
    :standard_name = "sea_ice_area_fraction";
    :units = "%";
    :coordinates = "lon lat";
    :grid_mapping = "crs";
    :source = "DMI";
    :_FillValue = -99S; // short
    :scale_factor = 1.0f; // float
    :add_offset = 0.0f; // float
    :_ChunkSizes = 1U, 1659U, 1116U; // uint

  short concentration_range(time=1, yc=3318, xc=2232);
    :long_name = "concentration range";
    :units = "%";
    :coordinates = "lon lat";
```

```
  :grid_mapping = "crs";
  :_FillValue = -99S; // short
  :comments = "Range of the analyized ice consentration value";
  :_ChunkSizes = 1U, 1659U, 1116U; // uint

short ice_poly_id_grid(time=1, yc=3318, xc=2232);
  :_FillValue = -99S; // short
  :long_name = "Reference ids for rasterized sea-ice-polygons";
  :units = "1";
  :coordinates = "lon lat";
  :grid_mapping = "crs";
  :source = "DMI";
  :scale_factor = 1.0; // double
  :add_offset = 0.0; // double
  :_ChunkSizes = 1U, 1659U, 1116U; // uint

short polygon_id(time=1, polygon_reference=1000);
  :long_name = "Ids corresponding to sea-ice-polygon reference ids";
  :units = "1";
  :source = "DMI";
  :nodata_value = "-9";
  :_ChunkSizes = 1U, 1000U; // uint

short CT(time=1, polygon_reference=1000);
  :long_name = "Total concentration (SIGRID3-code)";
  :standard_name = "sea_ice_area_fraction";
  :units = "1";
  :source = "DMI";
  :nodata_value = "-9";
  :_ChunkSizes = 1U, 1000U; // uint

short CA(time=1, polygon_reference=1000);
  :source = "DMI";
  :nodata_value = "-9";
  :units = "1";
  :long_name = "Partial concentration of thickest ice (SIGRID3-code)";
  :standard_name = "sea_ice_area_fraction";
  :_ChunkSizes = 1U, 1000U; // uint

short SA(time=1, polygon_reference=1000);
  :long_name = "Stage of development of thickest ice (SIGRID3-code)";
  :standard_name = "sea_ice_classification";
  :units = "1";
  :source = "DMI";
  :nodata_value = "-9";
  :_ChunkSizes = 1U, 1000U; // uint

short FA(time=1, polygon_reference=1000);
  :standard_name = "sea_ice_classification";
  :source = "DMI";
  :nodata_value = "-9";
```

```
  :long_name = "Form of thickest ice (SIGRID3-code)";
  :units = "1";
  :_ChunkSizes = 1U, 1000U; // uint

short CB(time=1, polygon_reference=1000);
  :long_name = "Partial concentration of second thickest ice (SIGRID3-code)";
  :units = "1";
  :source = "DMI";
  :nodata_value = "-9";
  :standard_name = "sea_ice_area_fraction";
  :_ChunkSizes = 1U, 1000U; // uint

short SB(time=1, polygon_reference=1000);
  :long_name = "Stage of development of second thickest Ice (SIGRID3-code)";
  :standard_name = "sea_ice_classification";
  :units = "1";
  :source = "DMI";
  :nodata_value = "-9";
  :_ChunkSizes = 1U, 1000U; // uint

short FB(time=1, polygon_reference=1000);
  :nodata_value = "-9";
  :standard_name = "sea_ice_classification";
  :units = "1";
  :source = "DMI";
  :long_name = "Form of second thickest ice (SIGRID3-code)";
  :_ChunkSizes = 1U, 1000U; // uint

short CC(time=1, polygon_reference=1000);
  :long_name = "Partial concentration of the third thickest ice (SIGRID3-code)";
  :standard_name = "sea_ice_area_fraction";
  :units = "1";
  :source = "DMI";
  :nodata_value = "-9";
  :_ChunkSizes = 1U, 1000U; // uint

short SC(time=1, polygon_reference=1000);
  :long_name = "Stage of development of third thickest ice (SIGRID3-code)";
  :standard_name = "sea_ice_classification";
  :units = "1";
  :source = "DMI";
  :nodata_value = "-9";
  :_ChunkSizes = 1U, 1000U; // uint

short FC(time=1, polygon_reference=1000);
  :long_name = "Form of third thickest ice (SIGRID3-code)";
  :standard_name = "sea_ice_classification";
  :source = "DMI";
  :nodata_value = "-9";
  :units = "1";
  :_ChunkSizes = 1U, 1000U; // uint
```

```
  short CN(time=1, polygon_reference=1000);
    :standard_name = "sea_ice_classification";
    :source = "DMI";
    :nodata_value = "-9";
    :long_name = "Stage of development of ice thicker than SA but with concentration less than 1/10 (SIGRID3-
code)";
    :units = "1";
    :_ChunkSizes = 1U, 1000U; // uint

  short CD(time=1, polygon_reference=1000);
    :long_name = "Stage of development of any remaining class of ice (SIGRID3-code)";
    :standard_name = "sea_ice_classification";
    :units = "1";
    :source = "DMI";
    :nodata_value = "-9";
    :_ChunkSizes = 1U, 1000U; // uint

  short CF(time=1, polygon_reference=1000);
    :standard_name = "sea_ice_classification";
    :units = "1";
    :source = "DMI";
    :nodata_value = "-9";
    :long_name = "Predominant and secondary forms of ice (SIGRID3-code)";
    :_ChunkSizes = 1U, 1000U; // uint

  char POLY_TYPE(time=1, polygon_reference=1000);
    :long_name = "General surface type: I=Ice, W=Water, (L=Land - not used by DMI)";
    :source = "DMI";
    :nodata_value = "N";
    :_ChunkSizes = 1U, 1000U; // uint

 // global attributes:
 :title = "Arctic Sea Ice Greenland Overview";
 :Conventions = "CF-1.4";
 :netcdf_version_id = "4.6.0";
 :creation_date = "2020-09-08T21:20:16Z";
 :produced_date = "2020-09-06T12:00:00Z";
 :valid_date = "2020-09-06T12:00:00Z";
 :product_version = "1.0";
 :software_version = "1.0";
 :comment = "The ice concentration is based on a manual interpretation of different satellite data. For more
information about the the product contact the national ice service";
 :satellite = "NA";
 :sensor = "SAR";
 :spatial_resolution_latitude = 1000.0f; // float
 :spatial_resolution_longitude = 1000.0f; // float
 :southernmost_latitude = 54.629f; // float
 :northernmost_latitude = 85.239235f; // float
 :westernmost_longitude = -105.15334f; // float
 :easternmost_longitude = 22.95248f; // float
```

```
  :field_type = "daily";
  :institution = "DMI";
  :institution_references = "http://www.dmi.dk";
  :contact = "iskort@dmi.dk";
  :operational_status = "operational";
}
```

Greenland dataset: cmems_obs-si_arc_physic_nrt_1km-grl_P1D-irr, example:

```
netcdf file: ice_conc_greenland_202009082105.nc {
  dimensions:
    time = 1;
    yc = 3318;
    xc = 2232;
    polygon_reference = 1000;
  variables:
    int time(time=1);
      :long_name = "reference time of sea ice file";
      :units = "seconds since 1981-01-01 00:00:00";
      :calendar = "standard";

    float yc(yc=3318);
      :axis = "Y";
      :long_name = "y-coordinate in Cartesian system";
      :units = "m";

    float xc(xc=2232);
      :axis = "X";
      :long_name = "x-coordinate in Cartesian system";
      :units = "m";

    float lat(yc=3318, xc=2232);
      :long_name = "latitude";
      :units = "degrees_north";
      :_ChunkSizes = 1106U, 744U; // uint

    float lon(yc=3318, xc=2232);
      :long_name = "longitude";
      :units = "degrees_east";
      :_ChunkSizes = 1106U, 744U; // uint

    char crs;
      :straight_vertical_longitude_from_pole = -45L; // long
      :latitude_of_projection_origin = 90L; // long
      :standard_parallel = 90L; // long
      :grid_mapping_name = "polar_stereographic";
      :false_easting = 0.0f; // float
      :false_northing = 0.0f; // float
      :proj4_string = " +proj=stere +lon_0=-45 +lat_ts=90 +lat_0=90 +a=6371000 +b=6371000";
```

```
short ice_concentration(time=1, yc=3318, xc=2232);
  :long_name = "sea ice concentration";
  :standard_name = "sea_ice_area_fraction";
  :units = "%";
  :coordinates = "lon lat";
  :grid_mapping = "crs";
  :source = "DMI";
  :_FillValue = -99S; // short
  :scale_factor = 1.0f; // float
  :add_offset = 0.0f; // float
  :_ChunkSizes = 1U, 1659U, 1116U; // uint

short concentration_range(time=1, yc=3318, xc=2232);
  :long_name = "concentration range";
  :units = "%";
  :coordinates = "lon lat";
  :grid_mapping = "crs";
  :_FillValue = -99S; // short
  :comments = "Range of the analyized ice consentration value";
  :_ChunkSizes = 1U, 1659U, 1116U; // uint

short ice_poly_id_grid(time=1, yc=3318, xc=2232);
  :_FillValue = -99S; // short
  :long_name = "Reference ids for rasterized sea-ice-polygons";
  :units = "1";
  :coordinates = "lon lat";
  :grid_mapping = "crs";
  :source = "DMI";
  :scale_factor = 1.0; // double
  :add_offset = 0.0; // double
  :_ChunkSizes = 1U, 1659U, 1116U; // uint

short polygon_id(time=1, polygon_reference=1000);
  :long_name = "Ids corresponding to sea-ice-polygon reference ids";
  :units = "1";
  :source = "DMI";
  :nodata_value = "-9";
  :_ChunkSizes = 1U, 1000U; // uint

short CT(time=1, polygon_reference=1000);
  :long_name = "Total concentration (SIGRID3-code)";
  :standard_name = "sea_ice_area_fraction";
  :units = "1";
  :source = "DMI";
  :nodata_value = "-9";
  :_ChunkSizes = 1U, 1000U; // uint

short CA(time=1, polygon_reference=1000);
  :source = "DMI";
  :nodata_value = "-9";
  :units = "1";
```

```
  :long_name = "Partial concentration of thickest ice (SIGRID3-code)";
  :standard_name = "sea_ice_area_fraction";
  :_ChunkSizes = 1U, 1000U; // uint

short SA(time=1, polygon_reference=1000);
  :long_name = "Stage of development of thickest ice (SIGRID3-code)";
  :standard_name = "sea_ice_classification";
  :units = "1";
  :source = "DMI";
  :nodata_value = "-9";
  :_ChunkSizes = 1U, 1000U; // uint

short FA(time=1, polygon_reference=1000);
  :standard_name = "sea_ice_classification";
  :source = "DMI";
  :nodata_value = "-9";
  :long_name = "Form of thickest ice (SIGRID3-code)";
  :units = "1";
  :_ChunkSizes = 1U, 1000U; // uint

short CB(time=1, polygon_reference=1000);
  :long_name = "Partial concentration of second thickest ice (SIGRID3-code)";
  :units = "1";
  :source = "DMI";
  :nodata_value = "-9";
  :standard_name = "sea_ice_area_fraction";
  :_ChunkSizes = 1U, 1000U; // uint

short SB(time=1, polygon_reference=1000);
  :long_name = "Stage of development of second thickest Ice (SIGRID3-code)";
  :standard_name = "sea_ice_classification";
  :units = "1";
  :source = "DMI";
  :nodata_value = "-9";
  :_ChunkSizes = 1U, 1000U; // uint

short FB(time=1, polygon_reference=1000);
  :nodata_value = "-9";
  :standard_name = "sea_ice_classification";
  :units = "1";
  :source = "DMI";
  :long_name = "Form of second thickest ice (SIGRID3-code)";
  :_ChunkSizes = 1U, 1000U; // uint

short CC(time=1, polygon_reference=1000);
  :long_name = "Partial concentration of the third thickest ice (SIGRID3-code)";
  :standard_name = "sea_ice_area_fraction";
  :units = "1";
  :source = "DMI";
  :nodata_value = "-9";
  :_ChunkSizes = 1U, 1000U; // uint
```

```
  short SC(time=1, polygon_reference=1000);
    :long_name = "Stage of development of third thickest ice (SIGRID3-code)";
    :standard_name = "sea_ice_classification";
    :units = "1";
    :source = "DMI";
    :nodata_value = "-9";
    :_ChunkSizes = 1U, 1000U; // uint

  short FC(time=1, polygon_reference=1000);
    :long_name = "Form of third thickest ice (SIGRID3-code)";
    :standard_name = "sea_ice_classification";
    :source = "DMI";
    :nodata_value = "-9";
    :units = "1";
    :_ChunkSizes = 1U, 1000U; // uint

  short CN(time=1, polygon_reference=1000);
    :standard_name = "sea_ice_classification";
    :source = "DMI";
    :nodata_value = "-9";
    :long_name = "Stage of development of ice thicker than SA but with concentration less than 1/10 (SIGRID3-
code)";
    :units = "1";
    :_ChunkSizes = 1U, 1000U; // uint

  short CD(time=1, polygon_reference=1000);
    :long_name = "Stage of development of any remaining class of ice (SIGRID3-code)";
    :standard_name = "sea_ice_classification";
    :units = "1";
    :source = "DMI";
    :nodata_value = "-9";
    :_ChunkSizes = 1U, 1000U; // uint

  short CF(time=1, polygon_reference=1000);
    :standard_name = "sea_ice_classification";
    :units = "1";
    :source = "DMI";
    :nodata_value = "-9";
    :long_name = "Predominant and secondary forms of ice (SIGRID3-code)";
    :_ChunkSizes = 1U, 1000U; // uint

  char POLY_TYPE(time=1, polygon_reference=1000);
    :long_name = "General surface type: I=Ice, W=Water, (L=Land - not used by DMI)";
    :source = "DMI";
    :nodata_value = "N";
    :_ChunkSizes = 1U, 1000U; // uint

 // global attributes:
 :title = "Arctic Sea Ice Greenland";
 :Conventions = "CF-1.4";
```

```
  :netcdf_version_id = "4.6.0";
  :creation_date = "2020-09-09T09:00:06Z";
  :produced_date = "2020-09-08T21:05:00Z";
  :valid_date = "2020-09-08T21:05:00Z";
  :product_version = "1.0";
  :software_version = "1.0";
  :comment = "The ice concentration is based on a manual interpretation of different satellite data. For more
information about the the product contact the national ice service";
  :satellite = "NA";
  :sensor = "SAR";
  :spatial_resolution_latitude = 1000.0f; // float
  :spatial_resolution_longitude = 1000.0f; // float
  :southernmost_latitude = 54.629f; // float
  :northernmost_latitude = 85.239235f; // float
  :westernmost_longitude = -105.15334f; // float
  :easternmost_longitude = 22.95248f; // float
  :field_type = "daily";
  :institution = "DMI";
  :institution_references = "http://www.dmi.dk";
  :contact = "iskort@dmi.dk";
  :operational_status = "operational";
}
```

Greenland dataset: DMI-ARC-SEAICE_CONC-L4-NRT-OBS, example:

```
netcdf ice_conc_greenland_201103280100 {
dimensions:
      time = 1 ;
      xc = 2232 ;
      yc = 3318 ;
variables:
      int time(time) ;
            time:long_name = "reference time of sea ice file" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
      float yc(yc) ;
            yc:axis = "Y" ;
            yc:long_name = "y-coordinate in Cartesian system" ;
            yc:units = "m" ;
      float xc(xc) ;
            xc:axis = "X" ;
            xc:long_name = "x-coordinate in Cartesian system" ;
            xc:units = "m" ;
      float lat(yc, xc) ;
            lat:long_name = "latitude" ;
            lat:units = "degrees_north" ;
      float lon(yc, xc) ;
            lon:long_name = "longitude" ;
            lon:units = "degrees_east" ;
      char crs ;
            crs:grid_mapping_name = "polar_stereographic" ;
            crs:false_easting = 0.f ;
```

```
            crs:false_northing = 0.f ;
            crs:proj4_string = " +a=6371000 +b=6371000 +lat_ts=90 +lon_0=-45 +proj=stere +lat_0=90"
 ;
      short ice_concentration(time, yc, xc) ;
            ice_concentration:long_name = "sea ice concentration" ;
            ice_concentration:standard_name = "sea_ice_area_fraction" ;
            ice_concentration:units = "%" ;
            ice_concentration:coordinates = "lon lat" ;
            ice_concentration:grid_mapping = "crs" ;
            ice_concentration:source = "DMI" ;
            ice_concentration:_FillValue = -99s ;
            ice_concentration:scale_factor = 1.f ;
            ice_concentration:add_offset = 0.f ;
      short concentration_range(time, yc, xc) ;
            concentration_range:long_name = "concentration range" ;
            concentration_range:units = "%" ;
            concentration_range:coordinates = "lon lat" ;
            concentration_range:grid_mapping = "crs" ;
            concentration_range:_FillValue = -99s ;
            concentration_range:comments = "Range of the analyized ice consentration value" ;

// global attributes:
            :title = "Total Sea Ice Concentration from national ice services." ;
            :Conventions = "CF-1.4" ;
            :netcdf_version_id = "3.6.2-beta6" ;
            :creation_date = "2011-03-28T23:31:33Z" ;
            :produced_date = "2011-03-28T01:00:00Z" ;
            :valid_date = "2011-03-28T01:00:00Z" ;
            :product_version = "1.0" ;
            :software_version = "1.0" ;
            :comment = "The ice concentration is based on a manual interpretation of different sate
llite data. For more information about the the product contact the national ice service" ;
            :satellite = "NA" ;
            :sensor = "SAR" ;
            :spatial_resolution_latitude = 1000.f ;
            :spatial_resolution_longitude = 1000.f ;
            :southernmost_latitude = 54.629f ;
            :northernmost_latitude = 85.23923f ;
            :westernmost_longitude = -105.1533f ;
            :easternmost_longitude = 22.95248f ;
            :field_type = "daily" ;
            :institution = "DMI" ;
            :institution_references = "http://www.dmi.dk" ;
            :contact = "iskort@dmi.dk" ;
            :operational_status = "operational" ;
```

Greenland dataset: DMI-ARC-SEAICE_CONC_OVERVIEW-L4-NRT-OBS, example:

```
netcdf ice_conc_overview_greenland_201105221200 {
dimensions:
            time = 1 ;
            xc = 2232 ;
```

```
        yc = 3318 ;
variables:
        int time(time) ;
                time:long_name = "reference time of sea ice file" ;
                time:units = "seconds since 1981-01-01 00:00:00" ;
        float yc(yc) ;
                yc:axis = "Y" ;
                yc:long_name = "y-coordinate in Cartesian system" ;
                yc:units = "m" ;
        float xc(xc) ;
                xc:axis = "X" ;
                xc:long_name = "x-coordinate in Cartesian system" ;
                xc:units = "m" ;
        float lat(yc, xc) ;
                lat:long_name = "latitude" ;
                lat:units = "degrees_north" ;
        float lon(yc, xc) ;
                lon:long_name = "longitude" ;
                lon:units = "degrees_east" ;
        char crs ;
                crs:grid_mapping_name = "polar_stereographic" ;
                crs:false_easting = 0.f ;
                crs:false_northing = 0.f ;
                crs:proj4_string = " +a=6371000 +b=6371000 +lat_ts=90 +lon_0=-45 +proj=stere +lat_0=90" ;
        short ice_concentration(time, yc, xc) ;
                ice_concentration:long_name = "sea ice concentration" ;
                ice_concentration:standard_name = "sea_ice_area_fraction" ;
                ice_concentration:units = "%" ;
                ice_concentration:coordinates = "lon lat" ;
                ice_concentration:grid_mapping = "crs" ;
                ice_concentration:source = "DMI" ;
                ice_concentration:_FillValue = -99s ;
                ice_concentration:scale_factor = 1.f ;
                ice_concentration:add_offset = 0.f ;
        short concentration_range(time, yc, xc) ;
                concentration_range:long_name = "concentration range" ;
                concentration_range:units = "%" ;
                concentration_range:coordinates = "lon lat" ;
                concentration_range:grid_mapping = "crs" ;
                concentration_range:_FillValue = -99s ;
                concentration_range:comments = "Range of the analyized ice consentration value" ;

// global attributes:
                :title = "Total Sea Ice Concentration from national ice services." ;
                :Conventions = "CF-1.4" ;
                :netcdf_version_id = "3.6.2-beta6" ;
                :creation_date = "2011-05-23T23:16:47Z" ;
                :produced_date = "2011-05-22T12:00:00Z" ;
                :valid_date = "2011-05-22T12:00:00Z" ;
                :product_version = "1.0" ;
                :software_version = "1.0" ;
```

:comment = "The ice concentration is based on a manual interpretation of different satellite data. For more information about the the product contact the national ice service" ;
:satellite = "NA" ;
:sensor = "SAR" ;
:spatial_resolution_latitude = 1000.f ;
:spatial_resolution_longitude = 1000.f ;
:southernmost_latitude = 54.629f ;
:northernmost_latitude = 85.23923f ;
:westernmost_longitude = -105.1533f ;
:easternmost_longitude = 22.95248f ;
:field_type = "daily" ;
:institution = "DMI" ;
:institution_references = "http://www.dmi.dk" ;
:contact = "iskort@dmi.dk" ;
:operational_status = "operational" ;}